

Bluestock Mutual Fund Analytics Capstone Project

Executive Summary

The Bluestock Mutual Fund Analytics Project was developed to analyze mutual fund performance, investor behavior, industry trends, and portfolio risk using Python, SQL, and Power BI. The project involved building a complete analytics pipeline beginning with data ingestion and cleaning, followed by exploratory data analysis, performance evaluation, advanced risk analytics, and dashboard development. The solution provides insights into fund performance, SIP growth trends, investor demographics, risk-adjusted returns, and portfolio concentration. A Power BI dashboard was created to enable interactive analysis of mutual fund data. Advanced analytics such as VaR, CVaR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Cohort Analysis, and HHI Concentration were also implemented.

1. Introduction

Mutual funds have become one of the most popular investment vehicles for retail investors. With increasing participation in SIP investments and diversified fund offerings, investors require analytical tools to evaluate fund performance and risk.

The objective of this project is to build a complete mutual fund analytics platform that enables:

- Fund performance evaluation
- Investor behavior analysis
- Risk measurement
- Industry trend analysis
- Dashboard-based decision support

2. Project Objectives

The key objectives of the project were:

1. Build a robust ETL pipeline.
2. Clean and validate mutual fund datasets.
3. Create a SQLite data warehouse.
4. Perform exploratory data analysis.
5. Compute advanced fund performance metrics.

6. Develop a Power BI dashboard.
7. Implement advanced risk analytics.
8. Generate actionable investment insights.

3. Data Sources

The project used ten datasets:

1. Fund Master
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

Additional live NAV data was fetched through MFAPI.

4. ETL Architecture

The ETL process consisted of three stages:

Extraction

- CSV datasets loaded using Pandas.
- Live NAV data collected using MFAPI.

Transformation

- Missing value handling
- Date standardization
- Duplicate removal

- Data validation
- Data quality checks

Loading

Cleaned datasets were loaded into SQLite using SQLAlchemy.

Insert ETL workflow diagram here.

5. Data Warehouse Design

A star schema was designed with:

Dimension Tables:

- dim_fund
- dim_date

Fact Tables:

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

Primary and foreign keys were established using AMFI codes and date fields.

Insert schema diagram here.

6. Exploratory Data Analysis

➤ NAV Trend Analysis

Daily NAV values were analyzed across all mutual fund schemes.

Insert NAV Trend chart.

Observation:

Most schemes demonstrated consistent long-term growth with temporary corrections during volatile market periods.

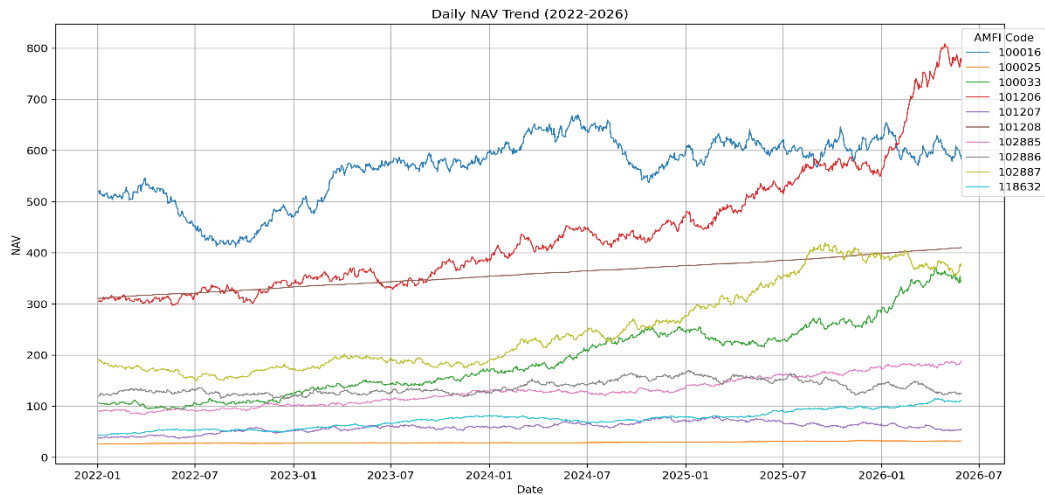


Fig: 6.1 NAV Trend Analysis

The NAV Trend Analysis illustrates the historical movement of Net Asset Values across mutual fund schemes between 2022 and 2026. The trend indicates a generally upward trajectory for most funds, reflecting long-term wealth creation and market growth. Periodic fluctuations observed during the timeline correspond to market volatility, economic events, and investor sentiment changes. Despite short-term corrections, most schemes demonstrated resilience and recovered over time, highlighting the benefits of long-term investing. This analysis helps investors understand fund stability and performance consistency across varying market conditions.

➤ AUM Growth Analysis

AUM growth was analyzed across fund houses.

Insert AUM Growth chart.

Observation: Large fund houses such as SBI, ICICI Prudential, and HDFC maintained dominant market share.

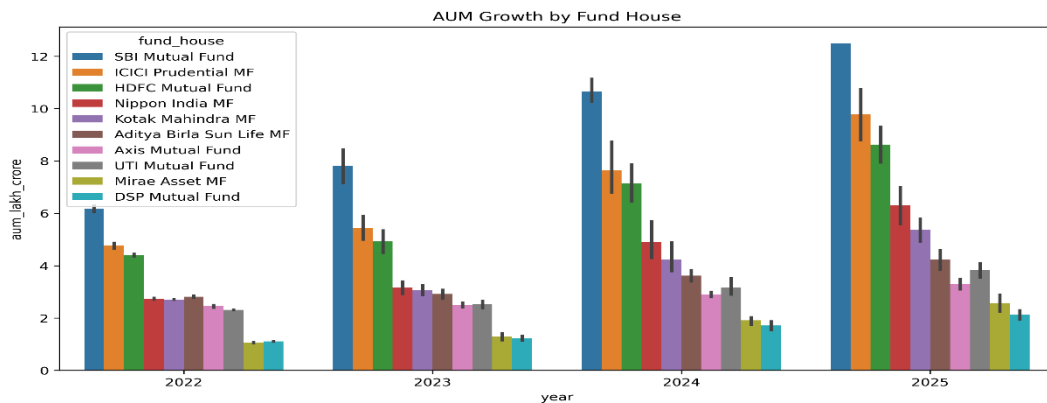


Fig: 6.2 AUM Growth Analysis

The Assets Under Management (AUM) Growth Analysis compares the total assets managed by different fund houses over the study period. The chart highlights the dominance of leading asset management companies such as SBI Mutual Fund, HDFC Mutual Fund, and ICICI Prudential Mutual Fund. Consistent AUM growth reflects investor confidence, strong fund performance, and effective fund management practices. The increasing AUM trend also indicates the growing popularity of mutual funds as an investment avenue in India.

➤ SIP Inflow Analysis

Monthly SIP inflows were studied between 2022 and 2025.

Insert SIP Trend chart.

Observation:

SIP inflows reached record highs, reflecting growing retail investor participation.

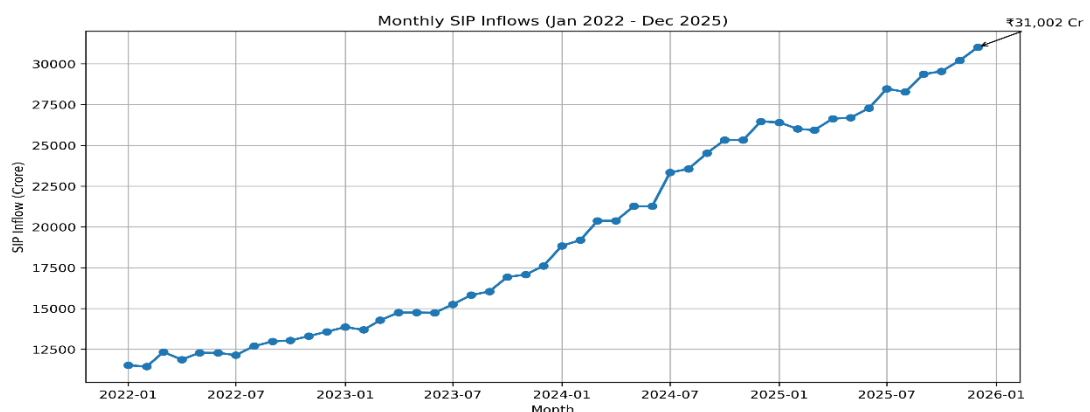


Fig: 6.3 SIP Inflow Analysis

The SIP Inflow Analysis examines monthly Systematic Investment Plan contributions from investors between 2022 and 2025. The trend demonstrates steady growth in SIP investments, culminating in record-high inflows during the later years of the study period. This increase reflects rising financial awareness, disciplined investment behavior, and greater participation of retail investors in capital markets. The findings indicate that SIPs continue to be a preferred investment method due to their affordability, convenience, and ability to mitigate market timing risks.

➤ Investor Demographics

Age group and geographic distributions were analyzed.

Insert demographic charts.

Observation:

Young and middle-aged investors contributed the largest share of SIP investments

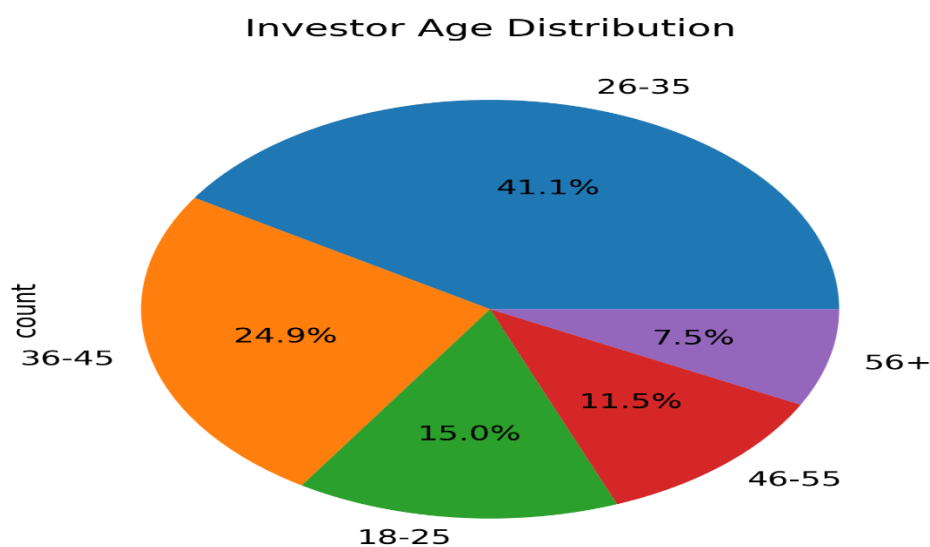


Fig: 6.4 Investor Demographics

The Investor Demographics Analysis provides insights into the characteristics of mutual fund investors based on age groups, gender distribution, and geographic location. The results reveal that young and middle-aged investors constitute the largest proportion of market participants, reflecting growing awareness of long-term financial planning. Geographic analysis shows that investment activity is concentrated in major urban and economically developed regions. Understanding investor demographics

enables asset management companies to design targeted products, improve customer engagement, and expand market penetration among underserved segments.

7. Performance Analytics

Several financial performance metrics were calculated.

- CAGR

Compound Annual Growth Rate was computed for 1-year, 3-year, and 5-year periods.

- Sharpe Ratio

Measured risk-adjusted returns.

- Sortino Ratio

Evaluated downside risk-adjusted returns.

- Alpha and Beta

Measured fund performance relative to benchmark indices.

- Maximum Drawdown

Identified worst historical declines.

	A	B	C	D	E	F	G
1	amfi_code	cagr_pct	sharpe_ratio	alpha	beta	max_drawdown	score
2	100016	2.635245812	-0.201517068	0.037475808	-0.058268431	-0.247344409	20.90032154
3	100025	4.455091156	-0.567094793	0.04281792	0.001158218	-0.043082638	9.646302251
4	100033	30.09970355	1.093698744	0.271953545	0.005103526	-0.162172086	89.06752412
5	101206	23.52048875	1.027213328	0.213997848	0.021086316	-0.112915963	75.24115756
6	101207	7.933121	0.162660769	0.108970919	-0.065289248	-0.354469165	45.17684887
7	101208	6.504429821	-0.815567368	0.060861458	0.00026671	-0.001622498	15.59485531
8	102885	18.21882609	0.817099337	0.170487568	-0.019486786	-0.108598622	58.68167203
9	102886	1.170938798	-0.205831608	0.028969398	-0.042125029	-0.280011236	16.88102894
10	102887	16.82867346	0.61951759	0.162112568	0.016682689	-0.215398271	58.03858521
11	118632	24.03119561	1.081659179	0.218293979	-0.008353832	-0.174140751	82.15434084
12	118633	16.262218	0.645207438	0.156438359	-0.036639007	-0.186297431	54.98392283
13	118634	16.36139648	0.448434009	0.175007217	0.103496637	-0.233448861	56.43086817
14	118635	15.94548119	0.664856999	0.151363546	-0.001433521	-0.116506245	49.8392283
15	118636	5.313155403	-0.356629508	0.05074773	0.001256674	-0.083163618	13.98713826
16	119092	6.363494758	0.030784851	0.068995298	0.009730745	-0.144016179	27.65273312
17	119093	7.891944369	0.129614498	0.082328013	0.025883212	-0.217513964	40.51446945
18	119094	28.19260822	0.998230987	0.260766691	-0.066265429	-0.209608839	84.5659164
19	119095	1.520455243	-0.075973571	0.048015954	-0.066951175	-0.516777537	23.47266881
20	119120	5.88527127	-0.226575387	0.056208952	-0.006413851	-0.043287289	16.07717042
21	119551	25.78492085	1.208266567	0.232010072	-0.031751441	-0.150123855	85.04823151
22	119552	21.50802179	0.953279497	0.198685994	-0.026159308	-0.118034935	68.64951768
23	119598	32.39808398	0.945308258	0.303369654	-0.023196062	-0.287060057	96.62379421
24	119599	2.053064146	-0.057186626	0.048824216	0.062001804	-0.525742212	26.20578778

Table:7.1 Top Funds Performance Comparison

Table 7.1 presents a comparative analysis of the top-performing mutual fund schemes based on key performance indicators. The evaluation incorporates long-term returns, risk-adjusted performance metrics, benchmark outperformance, market sensitivity, and downside risk exposure. Funds with higher Sharpe and Sortino ratios demonstrated

superior risk-adjusted returns, while positive Alpha values indicated the ability to outperform benchmark expectations. Beta values provided insight into market sensitivity, and Maximum Drawdown highlighted the extent of historical downside risk. This comparison enables investors to identify funds that balance return generation with effective risk management.

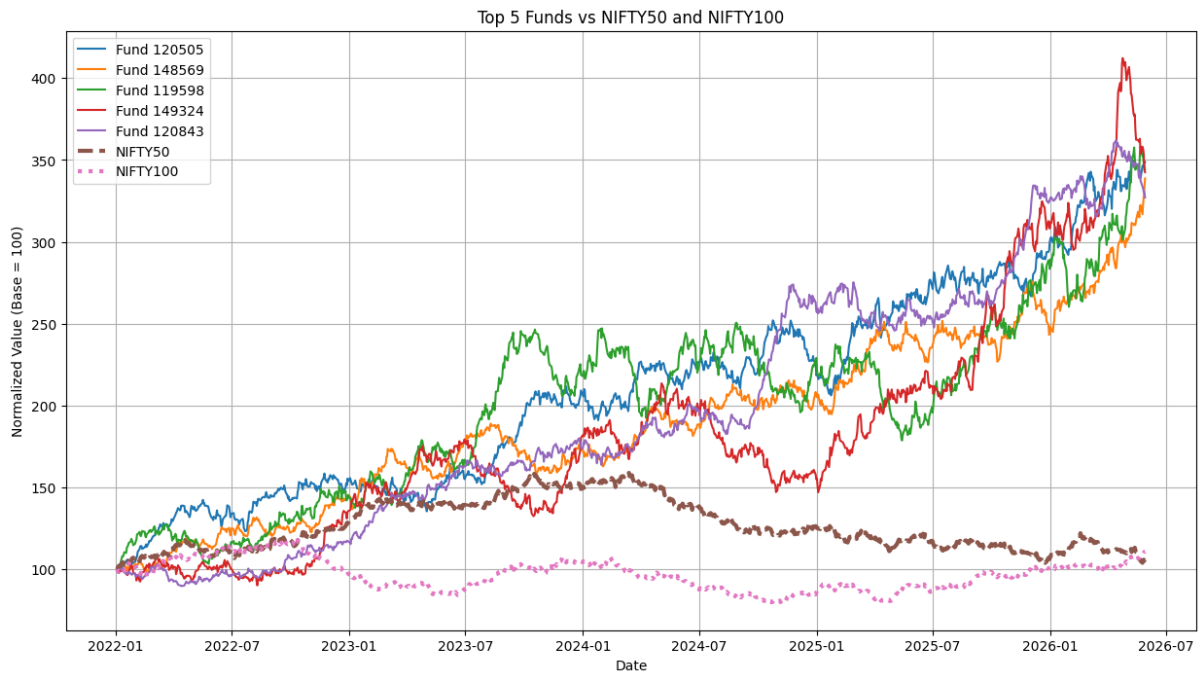


Figure 7.1: Rolling 90-Day Sharpe Ratio for Selected Funds

Figure 7.1 illustrates the Rolling 90-Day Sharpe Ratio for selected mutual fund schemes over the analysis period. The Sharpe Ratio measures risk-adjusted returns by comparing excess returns to portfolio volatility. Using a rolling 90-day window enables the observation of changes in fund performance consistency over time rather than relying on a single static value. Higher Sharpe Ratio values indicate superior risk-adjusted performance, while declining values suggest increased volatility or weaker returns. The chart reveals that certain funds maintained relatively stable Sharpe Ratios throughout the period, demonstrating consistent performance and effective risk management. In contrast, funds exhibiting sharp fluctuations were more sensitive to changing market conditions. This analysis provides investors with valuable insights into the stability and reliability of mutual fund performance across different market environments.

8. Advanced Analytics

Value at Risk (VaR)

95% historical VaR was computed for all schemes.

Conditional VaR (CVaR)

Expected losses beyond the VaR threshold were analyzed.

Rolling Sharpe Ratio

90-day rolling Sharpe Ratio trends were calculated.

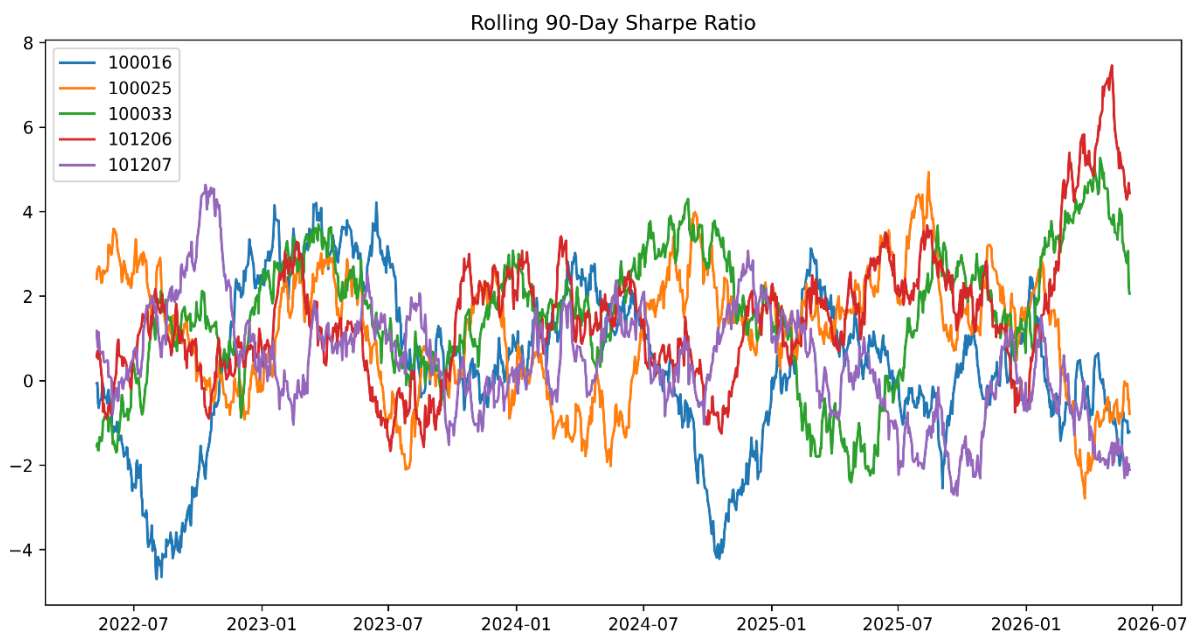


Fig:8.1 90-day rolling Sharpe Ratio

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Investor Cohort Analysis

Investor cohorts were grouped by first transaction year.

SIP Continuity Analysis

Investors with delayed SIP contributions were identified.

HHI Concentration Analysis

Portfolio concentration risk was measured using the Herfindahl-Hirschman Index.

9. Dashboard Development

A four-page Power BI dashboard was developed.

Page 1 – Industry Overview

Features:

- Total AUM

- SIP Inflows
- Folios
- Industry Trends

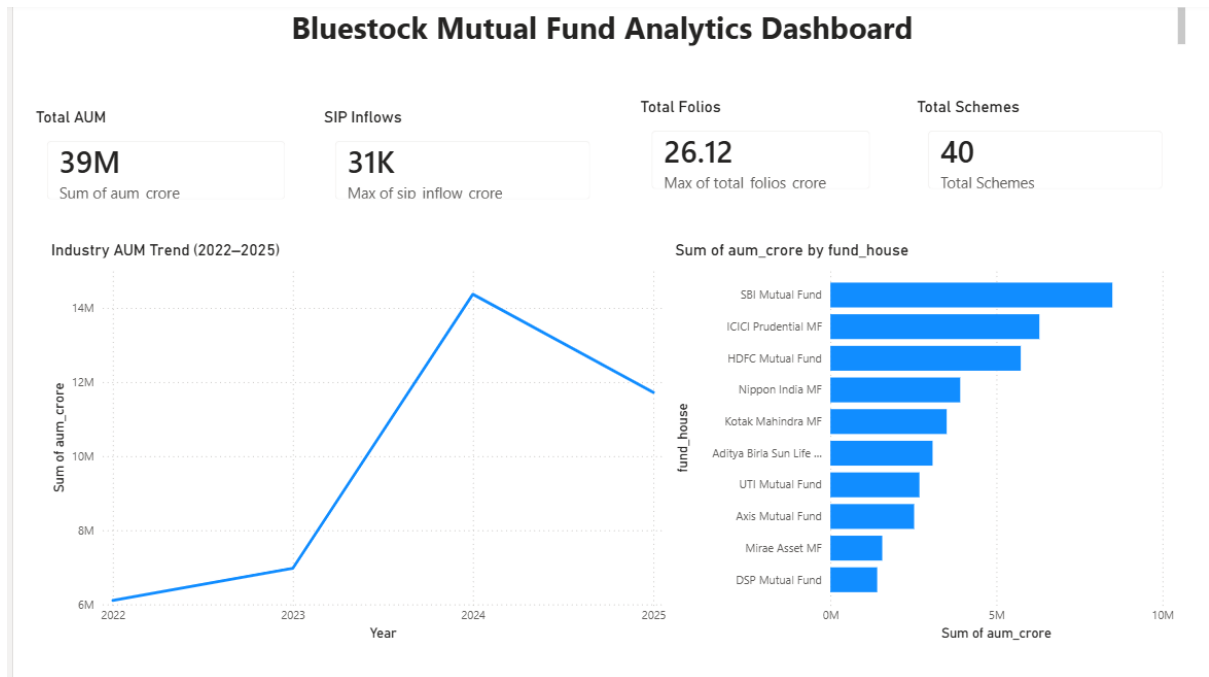


Fig:9.1 Industry Overview

The Industry Overview Dashboard provides a high-level summary of the mutual fund industry using key performance indicators and trend visualizations. The dashboard displays important metrics such as Total Assets Under Management (AUM), SIP Inflows, Total Folios, and the number of active schemes. Industry-wide trends are presented through AUM growth charts and fund house comparisons, enabling users to quickly assess market expansion and the relative position of major asset management companies. This dashboard serves as the primary entry point for understanding overall industry performance and investment activity.

Page 2 – Fund Performance

Features:

- Return vs Risk Analysis

- Fund Scorecard
- NAV Trends

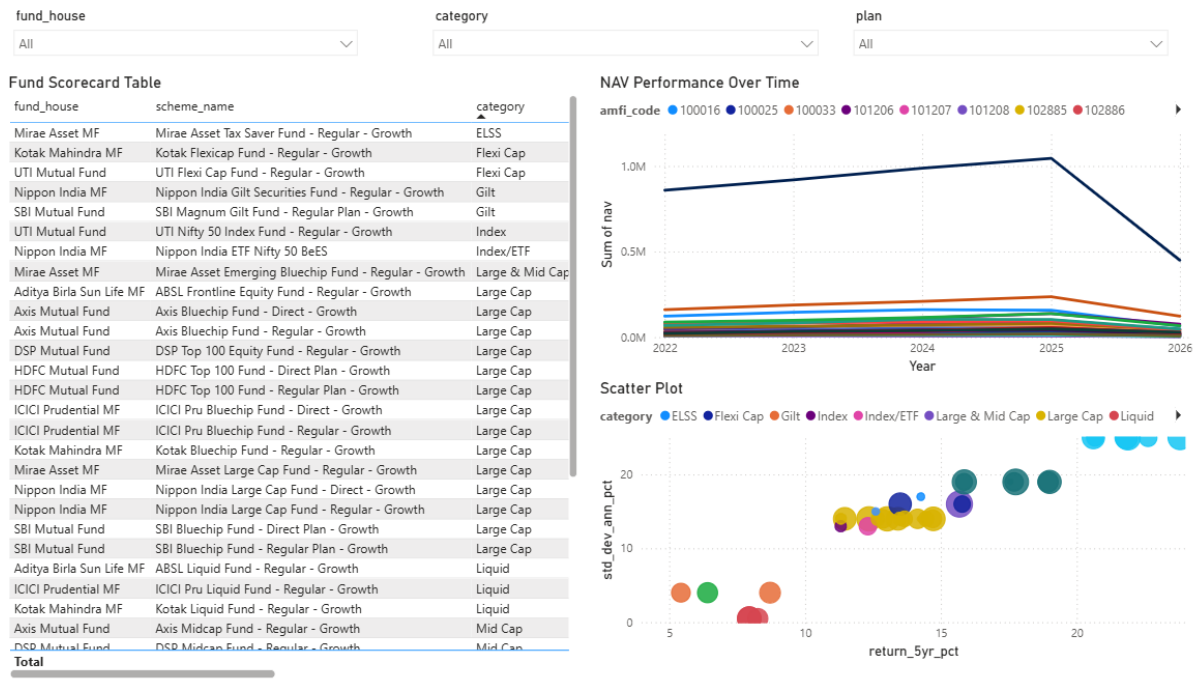


Fig:9.2 Fund Performance

The Fund Performance Dashboard focuses on evaluating mutual fund schemes based on returns, risk, and risk-adjusted performance metrics. Interactive visualizations allow users to compare funds using return-versus-risk scatter plots, fund scorecards, and NAV trend analysis. Key measures such as Sharpe Ratio, Alpha, Beta, CAGR, and Maximum Drawdown help investors identify high-performing funds while considering associated risks. Filters and slicers enable detailed exploration across fund houses, categories, and investment plans.

Page 3 – Investor Analytics

Features:

- State-wise Investments
- Transaction Types
- Investor Demographics

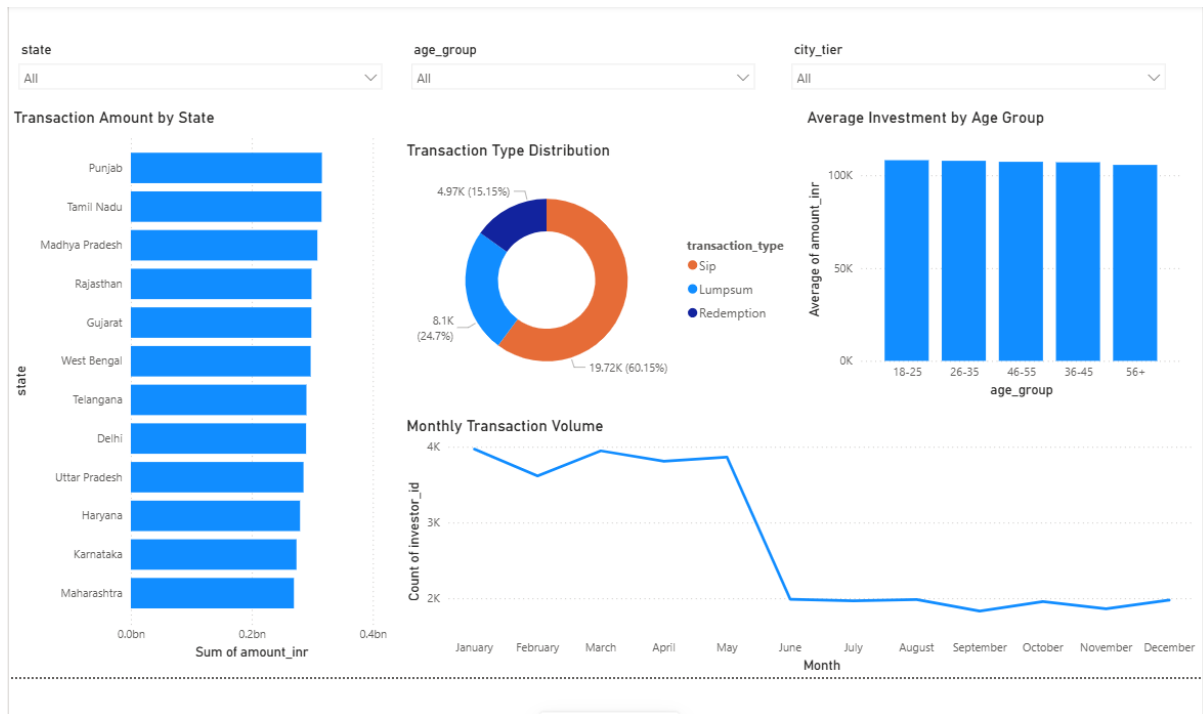


Fig:9.3 Investor Analytics

The Investor Analytics Dashboard provides insights into investor behavior, transaction patterns, and demographic characteristics. Visualizations include state-wise investment distribution, transaction type analysis, age-group participation, and transaction volume trends. The dashboard highlights how different investor segments contribute to mutual fund investments and helps identify geographic and demographic trends. These insights can support targeted marketing strategies and improve customer engagement initiatives for asset management companies.

Page 4 – SIP & Market Trends

Features:

- SIP vs Market Trends
- Category Inflows
- Market Comparisons

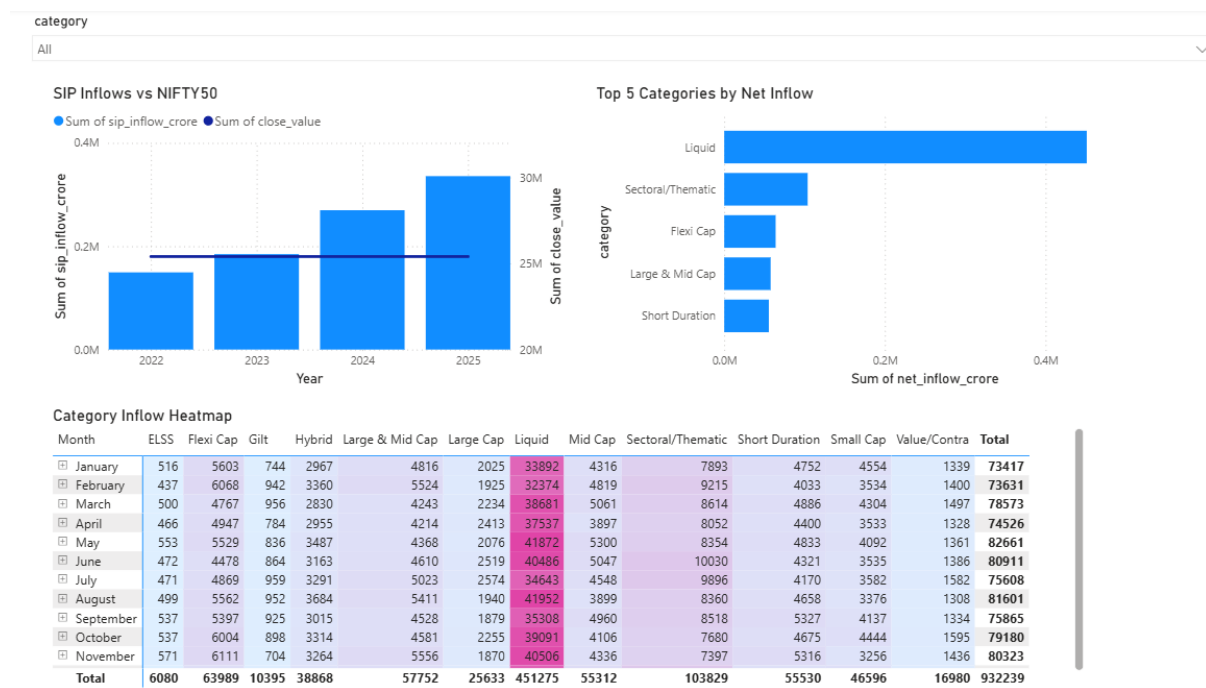


Fig:9.4 SIP & Market Trends

The SIP & Market Trends Dashboard examines the relationship between systematic investment activity and broader market movements. It presents SIP inflow trends alongside benchmark index performance, category-wise inflow analysis, and market participation patterns. The dashboard enables users to understand how investor sentiment and market conditions influence investment behavior over time. By combining SIP and market data, it provides a comprehensive view of industry growth, investment preferences, and emerging trends within the mutual fund ecosystem.

10. Key Findings

- SIP participation increased significantly during the study period.
- Large-cap funds delivered stable long-term returns.
- Risk-adjusted performance varied significantly across schemes.
- Some portfolios exhibited high concentration risk.
- Investor activity was concentrated in a few major states.
- Strong correlations existed among large-cap funds.
- High Sharpe Ratio funds consistently ranked higher in the fund scorecard.

11. Limitations

- Dataset limited to provided schemes.
- Historical analysis only.
- Recommendation engine uses rule-based logic.
- Market conditions may change future outcomes.

12. Recommendations

1. Integrate real-time market feeds.
2. Develop machine learning-based recommendation models.
3. Add portfolio optimization features.
4. Automate report generation.
5. Expand dashboard functionality.

13. Conclusion

The Bluestock Mutual Fund Analytics Project successfully demonstrated the complete lifecycle of a data analytics solution. From ETL and database design to dashboarding and advanced analytics, the project provided meaningful insights into mutual fund performance, investor behavior, and market trends. The resulting platform can assist investors, analysts, and fund managers in making informed decisions.

References

- [AMFI India](#)
- [MFAPI](#)
- [NSE India](#)
- [Power BI Documentation](#)
- [Pandas Documentation](#)
- [SQLAlchemy Documentation](#)
- [Python Documentation](#)